CAIR2 Integration — Quick Reference

ClaudMD / Aeliusmd | HL7 VXU v2.5.1 | SF-013259

1. Architecture

```
EMR publishes visit (EHRHeaders.IsPublish = 1)
EMR inserts EHRVaccineThirdPartySubmissions (SubmitStatus = 0)
        |
cair_service.py reads DB -> builds HL7 VXU -> sends SOAP -> updates status
```

2. Do We Need New Tables?

NO. Use existing ClaudMD tables only. Our script does not create cair_outbox or any new table.

3. Who INSERTs What (Our Script Never INSERTs)

Who	Table	When (INSERT)	Columns inserted by EMR
ClaudMD EMR	EHRVaccines	Nurse saves vaccination	CheckInId, ServiceCodeId, LotNumber, Dosage, VaccineDate, Manufacturer, Route, BodySite, IsVaccine
ClaudMD EMR	EHRHeaders	Provider publishes visit	CheckinId, IsPublish = 1
ClaudMD EMR	EHRVaccineThirdPartySubmissions	Visit ready for CAIR	EHRVaccineId, SubmitStatus = 0, AttemptCount = 0
Our Python script	Any table	Never (production worker)	No INSERT — read and UPDATE only
Dev seed script only	EHRVaccines + EHRVaccineThirdPartySubmissions	Testing in Sithum DB	INSERT test vaccines + queue rows — see Section 10

4. What Our Python Script Does (cair_service.py)

Step	Action	DB operation
1	Connect using .env — master DB (QA) or direct clinic DB (dev)	READ ClinicSetup OR CLINIC_DB_*
2	Get each clinic connection string	READ ClinicSetup columns: DatabaseServer, DatabaseName, DatabaseUser, DatabasePassword
3	Find pending CAIR submissions for published visits	READ EHRVaccineThirdPartySubmissions WHERE SubmitStatus IN (0,3) AND EHRHeaders.IsPublish = 1
4	Load patient and vaccine data to build HL7	READ EHRVaccines, Patients, CheckInsHeader, ServiceCodes (see Section 5)
5	Build HL7 VXU message	No DB write — uses CAIR PDF spec + Excel mapping
6	Send to CAIR2 SOAP endpoint	No DB write — submitSingleMessage (cair emails.txt)
7	On success — save result	UPDATE EHRVaccineThirdPartySubmissions + UPDATE EHRVaccines.IsSubmitted = 1
8	On failure — save error	UPDATE EHRVaccineThirdPartySubmissions (SubmitStatus, ErrorMessage, AttemptCount)

5. How Our Script Works (Detailed)

The background service is a single Python process: **cair_service.py**. It runs continuously on a server, wakes up every 5 minutes (configurable), checks all active clinics for pending CAIR submissions, and processes them. It does not run inside ClaudMD — it is a separate background worker.

5.1 Startup

```
1. Run:  python cair_service.py
2. Load settings from .env file (DB server, CAIR SOAP URL, org code SF-013259)
3. Create CairCoordinator
4. Enter infinite loop — repeat every SCHEDULER_INTERVAL_SECONDS (default 300 = 5 min)
```

5.2 Each Run Cycle (coordinator.run_once)

```
                         cair_service.py
                               |
                   CairCoordinator.run_once()
                               |
              Connect to master DB OR direct clinic DB (see Section 7)
              QA: READ ClinicSetup WHERE Active = 1
              Dev: use CLINIC_DB_* from .env (skip ClinicSetup)
                               |
              For each clinic: check if pending submissions exist
                               |
              ThreadPoolExecutor (3 workers by default)
                   /         |         \
             Clinic A     Clinic B     Clinic C
             (batch)      (batch)      (batch)
```

5.3 Per Clinic Batch (process_clinic_batch)

#	What happens	Python file / class
1	Connect to clinic DB using ClinicSetup credentials	cair_submission_repository.py
2	SELECT up to 100 pending submissions (SubmitStatus 0 or 3) where IsPublish=1	CairSubmissionRepository.get_eligible_submissions()
3	Set SubmitStatus = 4 (PROCESSING) on selected rows	CairSubmissionRepository.claim_submissions()
4	For each submission row — call process_submission()	processor.py — SubmissionProcessor

5.4 Per Submission Record (process_submission)

#	Step	What it does
1	Load data	JOIN EHRVaccines + Patients + CheckInsHeader + ServiceCodes + EHRHeaders
2	Build HL7	vxu_builder.py creates MSH, PID, PD1, ORC, RXA, OBX segments per CAIR PDF + Excel
3	Send SOAP	soap_client.py POSTs HL7 to CAIR stage/prod endpoint (submitSingleMessage)
4	Parse ACK	ack_parser.py reads MSA segment: AA=accept, AE=error, AR=reject
5a	If AA (success)	UPDATE submission SubmitStatus=1, save HL7+ACK, set EHRVaccines.IsSubmitted=1
5b	If AE (temp error)	UPDATE SubmitStatus=3 (RETRY), increment AttemptCount, wait and retry later
5c	If AR or max retries	UPDATE SubmitStatus=2 (FAILED), save ErrorMessage

5.5 Retry Logic

Attempt	Wait before retry	After 5 failures
1	5 minutes	
2	15 minutes	
3	30 minutes	
4	60 minutes	
5	120 minutes	Mark as FAILED (SubmitStatus = 2)

Retries are automatic. A failed row with SubmitStatus=3 is picked up again on the next run after the wait period. New pending rows (SubmitStatus=0) and retry rows share the same queue.

5.6 Multi-Clinic Rules

- One shared worker pool (default 3 threads) — not one worker per clinic forever
- Each worker handles one clinic batch (max 100 records), then moves to next clinic
- Only one active batch per clinic at a time (avoids duplicate sends)
- Clinics with no pending work are skipped

5.7 Script Components (File Map)

File	Role
cair_service.py	Main entry — infinite scheduler loop
cair_integration/config.py	Reads .env — DB credentials, CAIR URL, batch size
cair_integration/constants.py	SubmitStatus values (0–4)
worker/coordinator.py	Finds clinics with work, dispatches to thread pool
worker/processor.py	Processes one submission: load -> HL7 -> send -> update
worker/retry.py	Calculates retry delay (exponential backoff)
repository/cair_submission_repository.py	All SQL READ and UPDATE queries
hl7/vxu_builder.py	Builds HL7 VXU message string
cair/soap_client.py	Sends HL7 to CAIR SOAP endpoint
cair/ack_parser.py	Parses CAIR ACK response (AA/AE/AR)
demo_build_vxu.py	Test HL7 output without database connection
scripts/seed_sithum_cair_test_data.py	Dev only — insert test vaccines + queue rows in Sithum DB
run_dry_once.py	One worker cycle — read DB, build HL7, optional dry-run

5.8 Configuration (.env)

Two modes: QA uses **MASTER_DB_*** + ClinicSetup lookup. Development uses **CLINIC_DB_*** for a direct clinic connection (e.g. Sithum DB). When CLINIC_DB_* is set, ClinicSetup is skipped.

Setting	Example	Purpose
APP_ENV	development / qa	Environment label
CLINIC_DB_SERVER	10.103.0.211	Dev: direct clinic SQL Server
CLINIC_DB_NAME	ClaudMD_Development_Sithum	Dev: clinic database name
CLINIC_DB_USER / PASSWORD	testuser / ***	Dev: clinic DB login
CLINIC_NAME	Development Sithum	Dev: display name for logs
MASTER_DB_SERVER	10.103.0.201	QA: SQL Server for master DB
MASTER_DB_NAME	ClaudMD_QA_Setup	QA: master database name
DEFAULT_ACTIVATION_KEY	20000002	QA: clinic activation key
CAIR_SOAP_URL	stage endpoint	Where to send HL7 messages
CAIR_SOAP_USERNAME / PASSWORD	(from CAIR onboarding)	Required to actually send HL7
SENDING_FACILITY_ID	SF-013259	CAIR org code (MSH-4)
WORKER_BATCH_SIZE	100	Max submissions per clinic per run
WORKER_POOL_SIZE	3	Parallel clinic workers
SCHEDULER_INTERVAL_SECONDS	300	Seconds between each run cycle

Setting	Example	Purpose
MAX_RETRY_ATTEMPTS	5	Retries before marking FAILED

6. Columns Used Per Table

Below are the exact columns our script reads or updates in each table.

ClinicSetup (Master DB — ClaudMD_QA_Setup)

Column	Used for	Read / Write
ClinicID	Clinic identifier	READ
ClinicName	Display name	READ
DatabaseServer	SQL Server host for clinic DB	READ
DatabaseName	Clinic database name	READ
DatabaseUser	Clinic DB login	READ
DatabasePassword	Clinic DB password	READ
Active	Filter Active = 1	READ
ActivationKey	e.g. 20000002	READ

EHRHeaders (Clinic DB)

Column	HL7 / purpose	Read / Write
CheckinId	Links to EHRVaccines.CheckInId	READ
IsPublish	Must be 1 before CAIR send	READ (filter)
IsDeleted	Must be 0	READ (filter)

EHRVaccineThirdPartySubmissions (Clinic DB)

Column	Purpose	Read / Write
Id	Submission queue row ID	READ
EHRVaccineId	FK to EHRVaccines.Id	READ
SubmitStatus	0=pending, 1=success, 2=failed, 3=retry, 4=processing	READ + UPDATE
SubmittedDateTime	When CAIR accepted	UPDATE on success
LastAttemptDateTime	Last send attempt	UPDATE
AttemptCount	Retry counter	UPDATE
ExternalReferenceId	MSH-10 message control ID (GUID)	UPDATE on success
ErrorMessage	CAIR error or exception text	UPDATE on fail/retry
RequestPayload	HL7 VXU message sent	UPDATE
ResponsePayload	CAIR ACK response	UPDATE
IsDeleted	Must be 0	READ (filter)

EHRVaccines (Clinic DB)

Column	HL7 field (PDF / Excel)	Read / Write
Id	Internal vaccine record ID	READ

Column	HL7 field (PDF / Excel)	Read / Write
CheckInId	ORC-2/3 order number (Excel: checked_In.checkin_id)	READ
ServiceCodeId	Join to ServiceCodes	READ
LotNumber	RXA-15 substance lot number	READ
VaccineExpirationDate	RXA-16 expiration date	READ
Dosage	RXA-6 administered amount	READ
VaccineDate	RXA-3 administration date	READ
VaccineTime	RXA-3 administration time	READ
Manufacturer	RXA-17 manufacturer (need MVX code)	READ
Route	RXR-1 route	READ
BodySite	RXR-2 injection site	READ
IsVaccine	Filter IsVaccine = 1	READ (filter)
IsSubmitted	Flag after CAIR success	UPDATE = 1 on success
IsDeleted	Must be 0	READ (filter)

CheckInsHeader (Clinic DB)

Column	HL7 field (PDF / Excel)	Read / Write
Id	ORC-2/3 placer/filler order number	READ
PatientId	Join to Patients	READ
CheckInDate	Visit date (fallback for RXA-3)	READ
CheckInTime	Visit time (fallback for RXA-3)	READ
IsDeleted	Must be 0	READ (filter)

Patients (Clinic DB)

Column	HL7 field (PDF / Excel)	Read / Write
AccountNumber	PID-3 patient ID (Excel: Patient.Acc_no)	READ
LastName	PID-5 patient name	READ
FirstName	PID-5 patient name	READ
Initials	PID-5 middle name	READ
DateOfBirth	PID-7 date of birth	READ
GenderId	PID-8 sex (lookup M/F/X/U)	READ
HomePhone	PID-13 home phone (required per email)	READ
CellPhone	PID-13 cell phone (required per email)	READ
Email	PID-13 email (required per email)	READ
Address1	PID-11 address	READ
City	PID-11 city	READ
State	PID-11 state	READ
ZipCode	PID-11 zip	READ
IsDeleted	Must be 0	READ (filter)

ServiceCodes (Clinic DB)

Column	HL7 field (PDF / Excel)	Read / Write
Id	Join from EHRVaccines.ServiceCodeId	READ
NDCNumber	RXA-5 vaccine code (Excel: Charge_rec_detail.NDC_number)	READ
Description	RXA-5 vaccine description text	READ
CPTCode	Billing only — not primary CAIR field	READ (optional)
CvxCode	RXA-5 CVX code — NOT in DB yet, EMR must add	READ (future)
MvxCode	RXA-17 MVX code — NOT in DB yet, EMR must add	READ (future)

7. Database Connection

7.1 QA Mode (Master DB + ClinicSetup)

```
STEP A — Master DB (from .env):
  Server: 10.103.0.201
  Database: ClaudMD_QA_Setup
  READ dbo.ClinicSetup WHERE Active = 1

STEP B — Clinic DB (per ClinicSetup row):
  Example: ClaudMD_VCOMC_QA_2 (activation key 20000002)
  READ   -> build HL7 from tables in Section 6
  UPDATE -> EHRVaccineThirdPartySubmissions + EHRVaccines only
```

7.2 Development Mode (Direct Clinic DB — Sithum)

The Sithum development database is a **clinic DB**, not registered in ClinicSetup. Set **CLINIC_DB_*** in .env and leave MASTER_DB_* commented out. CAIR-related tables were synced from QA_2 — same 6 tables and columns.

```
Direct connection (from .env):
  Server:   10.103.0.211
  Database: ClaudMD_Development_Sithum
  User:     testuser

Tables verified (match QA_2):
  EHRHeaders, EHRVaccines, EHRVaccineThirdPartySubmissions,
  CheckInsHeader, Patients, ServiceCodes
```

SubmitStatus values (EHRVaccineThirdPartySubmissions.SubmitStatus)

Value	Name	Set by
0	PENDING	EMR on INSERT
1	SUCCESS	Our script on CAIR ACK = AA
2	FAILED	Our script on permanent error
3	RETRY	Our script on temporary error
4	PROCESSING	Our script while sending

8. HL7 Field Quick Map (PDF + Excel)

HL7	Source column	Spec reference
MSH-4	.env SENDING_FACILITY_ID = SF-013259	CAIR PDF + onboarding email
MSH-6	CAIR2	cair emails.txt sample
MSH-9	VXU^V04^VXU_V04	CAIR PDF
MSH-15/16	AL / AL	cair emails.txt sample
PID-3	Patients.AccountNumber	Excel mapping
PID-13	Patients.CellPhone, HomePhone, Email	cair emails.txt — required
ORC-2/3	CheckInsHeader.Id	Excel mapping
RXA-5	ServiceCodes.NDCNumber or CVX	Excel + CAIR PDF
RXA-15	EHRVaccines.LotNumber	Excel mapping

HL7	Source column	Spec reference
OBX-5	V01 Not VFC eligible	cair emails.txt sample

9. CAIR Endpoints & Commands

Item	Value
Stage endpoint	cdph-interop-stage.cdph.ca.gov/.../client_ServiceHttpSoap12Endpoint
SOAP operation	submitSingleMessage
Org code	SF-013259
Test HL7 (no DB)	python demo_build_vxu.py
Dry run (read DB, no send)	python run_dry_once.py
Run worker	python cair_service.py

10. Development Testing (Sithum DB + Seed Script)

Use **scripts/seed_sithum_cair_test_data.py** to insert test records in the development clinic DB before running the CAIR worker. **Seeding does NOT send HL7** — it only creates queue rows (SubmitStatus = 0). HL7 is built and sent only when **cair_service.py** or **run_dry_once.py** runs.

10.1 Seed Script Commands

Command	What it does
python scripts/seed_sithum_cair_test_data.py list	Show published visits, vaccine codes with NDC, existing submissions
python scripts/seed_sithum_cair_test_data.py seed	Create up to 3 test vaccines + queue rows (default --count 3)
python scripts/seed_sithum_cair_test_data.py seed --dry-run	Preview actions without writing to the database
python scripts/seed_sithum_cair_test_data.py verify	Confirm records are eligible for the CAIR worker (read-only)
python scripts/seed_sithum_cair_test_data.py reset	Reset LOT-CAIR-TEST-* submissions back to PENDING for re-testing

10.2 What the Seed Script Inserts

Table	Action	Details
EHRVaccines	INSERT or UPDATE	LotNumber, VaccineDate, Manufacturer, Route, BodySite, IsVaccine=1, ServiceCodeId with NDC
EHRVaccineThirdPartySubmissions	INSERT	EHRVaccineId, SubmitStatus=0 (PENDING), AttemptCount=0
EHRHeaders	No change	Uses existing published visits (IsPublish = 1)
Patients / CheckInsHeader	No change	Uses existing patient demographics

The script auto-discovers published visits where the patient has phone and/or email (PID-13 requirement) and picks vaccine ServiceCodes that have an NDC number. Test lot numbers use prefix **LOT-CAIR-TEST-**. Safe to re-run — skips vaccines that already have a submission row.

10.3 Testing Workflow (Step by Step)

```
1. Set CLINIC_DB_* in .env (Sithum development DB)
2. python scripts/seed_sithum_cair_test_data.py seed
3. python scripts/seed_sithum_cair_test_data.py verify
      -> SubmitStatus should still be 0, RequestPayload empty (no HL7 sent yet)
4. python run_dry_once.py
      -> builds HL7 from DB; does not send to CAIR unless configured
5. Add CAIR_SOAP_USERNAME and CAIR_SOAP_PASSWORD to .env
6. python cair_service.py
      -> sends HL7 to CAIR stage; updates SubmitStatus + saves RequestPayload
```

10.4 How to Tell If HL7 Was Sent

Check	Not sent yet	Sent / processed
SubmitStatus	0 (PENDING)	1=success, 2=failed, 3=retry, 4=processing
RequestPayload	NULL / empty	Contains HL7 VXU message text
SubmittedDateTime	NULL	Set on CAIR success (ACK = AA)
ResponsePayload	NULL	Contains CAIR ACK response

Note: CAIR_SOAP_USERNAME and CAIR_SOAP_PASSWORD must be set in .env before the worker can send to CAIR. Without credentials, only dry-run / verify steps work.